

Introduction to Probability Theory

PETER NYANDIRI

Strathmore Institute of Mathematical Sciences

Lecture 1: Introduction

Introduction

Definition (Statistics)

Statistics is the science of conducting studies to collect, organize, summarize, analyze, and draw conclusions from data.

Why learn Statistics

- To summarize business data
- To draw conclusion from data
- To make reliable forecast about business activities
- To improve business process (quality control)

Key Definitions

- ◇ **Population** is a set of all elements of interest in a particular study.
- ◇ **Sample** is a subset of the population.
- ◇ **Parameter** A numerical measure that describes a characteristic of a population.
- ◇ **Statistics** A numerical measure that describes a characteristic of a sample.
- ◇ **Variable** Is a characteristic or attribute that can assume different values.

Types of statistics

Descriptive statistics

- * Consists of the collection, organization, summarization and presentation of data.
- * E.g. tabular, graphical, and summary statistics

Inferential statistics

- * The branch of statistics that analyzes sample data to draw conclusions about a population.
- * E.g. estimation and hypothesis testing statistics

Types of Statistics

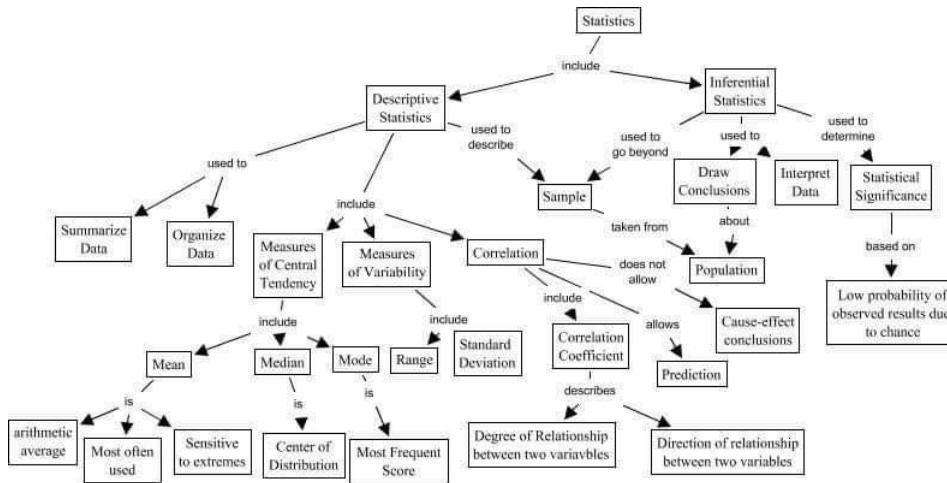


Figure: A summary of Statistics analysis

Data and Data sets

- ◇ **Data** are the facts and figures collected, summarized, analyzed, and interpreted
- ◇ **Dataset** is a collection of data.
- ◇ **Classification of data**
 - * Based on nature
 - * Based on source

Classification of Data

Based on nature

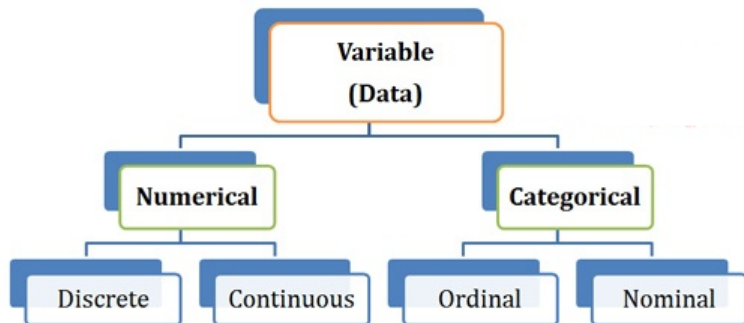


Figure: Data types

Based on nature

Definition (Qualitative data(Categorical data))

Are variables that can be placed into distinct categories, according to some characteristic or attribute.

- ◇ **Nominal data:** No natural ordering of the categories.
 - * political party (Democratic, Republican, Independent, etc.)
 - * religion (Christianity, Judaism, Islam, etc.)
- ◇ **Ordinal data:** There is natural ordering of the categories
 - * Evaluation of a new security system (very good, good, poor).
 - * Product quality rating (1: poor; 2: average; 3: good)

Based on nature

Definition (Quantitative data (Numerical))

Measures that take numerical values.

- ◇ **Discrete data:** Assume values that can be counted.
 - * Number of computers
 - * Number of cars

- ◇ **Continuous data:** Assume values that can be measured.
 - * height of a person
 - * Distance in KM

Classification of Data-Based on source

Definition (Primary Data)

Data that has been collected from first-hand-experience

Sources

- Direct Observations
- Experiments - Independent & Dependent variables.
- Surveys - Questionnaires & Interviews

Definition (Secondary Data)

Data collected from a source that has already been published

Sources

- Books
- Electronic Sources
- Magazines & Newspapers

Measurement of scales

Measurement scales are used to categorize and/or quantify variables.

Definition (Properties of Measurement Scales)

- ◇ **Identity:** Each value on the measurement scale has a unique meaning.
- ◇ **Magnitude:** Values on the measurement scale have an ordered relationship to one another. That is, some values are larger and some are smaller.
- ◇ **Equal intervals:** Scale units along the scale are equal to one another. This means, for example, that the difference between 1 and 2 would be equal to the difference between 19 and 20.
- ◇ **Absolute zero:** The scale has a true zero point, below which no values exist.

Types of Measures of Scales

Definition (Nominal Scale of Measurement)

- ◇ Only satisfies the identity property of measurement.
- ◇ Classifies the data into categories which no order or ranking can be imposed on the data.
- ◇ **Examples:**
 - i. Gender (male or female)
 - ii. Cell phones (Nokia, Samsung and Apple).

Definition (Ordinal Scale of Measurement)

- ◇ Has the property of both identity and magnitude.
- ◇ data that can be ordered; the differences cannot be measured
- ◇ **Examples:**
 - i. Grades in an exam (A, B, C, D)
 - ii. Product quality rating (1: poor; 2: average; 3: good)

Types of Measures of Scales

Definition (Interval Scale of Measurement)

- ◇ Has the properties of identity, magnitude, and equal intervals.
- ◇ Data with a definite ordering but no starting point; the differences can be measured.
- ◇ **Examples:**
 - i. Temperature

Definition (Ratio Scale of Measurement)

- ◇ Satisfies all four of the properties of measurement: identity, magnitude, equal intervals, and an absolute zero
- ◇ Data with a starting point that can be ordered; the differences have meaning and ratios can be calculated.
- ◇ **Examples:**
 - i. Height, Age, Weight

Sampling

Sampling is a process used in statistical analysis for selecting a sample from the population.

Sampling Techniques

- ◇ **Probability Sampling:** Every item in a population of interest, has a known, and usually equal chance of being chosen for inclusion in the sample. Example:
 - i. Simple random sampling
 - ii. Stratified random sampling
 - iii. Cluster random sampling
 - iv. Systematic random sampling
- ◇ **Non-probability Sampling:** Sample does not have known probability of being selected.
Example:
 - i. Convenience sampling
 - ii. Judgmental sampling

Probability Sampling

- ◇ **Simple Random Sampling (SRS):** All items in the population have equal chance of being selected.
- ◇ **Stratified Random Sampling:** Divide the population into groups called strata and then take a sample from each stratum).
- ◇ **Cluster Random Sampling:** Divide the population into clusters and then randomly select some of the clusters. All the members from these clusters are in the cluster sample.
- ◇ **Systematic Random Sampling:** randomly select a starting point and take every n -th piece of data from a listing of the population.

Non Probability Sampling

- ◇ **Convenience Sampling:** A convenience sample is made up of measurements, or observations that are most easily accessible.
- ◇ **Judgmental Sampling:** one in which an individual selects the items to be included depending on his judgment in the sample.

THANK YOU

QUESTIONS